

Leave-one-out disturbance ablation

Which single simplification of the full disturbance set changes policy success rate, measured on three manipulation tasks at six demonstration budgets

scope	3 tasks x 6 arms x 6 demonstration budgets = 108 trained policies
evidence	21,600 scored episodes (200 per cell), plus 21,600 discarded warm-up episodes
question	SR(full set minus one axis) - SR(full set), at a matched demonstration budget
contents	measured surface, leave-one-out deltas, within-cell dispersion, T2 milestone instrumentation, time-to-success, generation and training provenance, implementation flaws and open items
excluded	no interpretation, no ranking, no comparison against any other study or previously published surface
generated	2026-09-20 from repository revision 5e265ff

1 Design and definitions

The full disturbance set is $L3b = \{A, B, C\}$. Each leave-one-out arm removes exactly one axis from that set and keeps the other two. Every delta reported here is $SR(\text{arm with one axis removed}) - SR(L3b)$ at the same demonstration budget, so a positive delta means the simplified arm scored higher.

arm	disturbances active	leave-one-out meaning	sub-environments	eval env per slice s
L0	none	--	1 flat	<TASK>-L0
L1	A	--	1 flat	<TASK>-L1
L2	A, B	$L3b \setminus C$ (variant axis removed)	1 flat	<TASK>-L2
L3b	A, B, C (full set)	reference for every delta	10 variants	<TASK>-L3v0s
AC	A, C	$L3b \setminus B$ (goal/fixture pose removed)	10 variants	<TASK>-ACv0s
BC	B, C	$L3b \setminus A$ (manipulandum pose removed)	10 variants	<TASK>-BCv0s

1.1 What each axis randomises

axis	T1 cup_place	T2 drawer_stow	T3 push_target
A manipulandum pose	cup XY over 30x40 cm, yaw +/-90 deg	stow-object XY, yaw +/-45 deg	puck XY over 12x12 cm
B goal / fixture pose	goal disk XY over 20x20 cm	cabinet XY +/-5 cm, yaw +/-7.5 deg	target bearing +/-25 deg
C object variant	10 cup variants (2 sizes x 5 colours)	10 box variants	10 puck variants (radius / height)

1.2 Evaluation protocol

property	value
episodes scored per cell	200 (10 slices x 20 parallel environments)
slices per cell	10 independent processes, one scored batch each
warm-up	1 unscored batch per slice, reset seed 4900+s, outcomes discarded
scored reset seed	5000+s for slice s = 0..9
variant arms	diagonal: slice s evaluates sub-environment v0s
checkpoint	step 080000 of each cell, deterministic action selection
episode cap	T1 600 steps, T2 1,200 steps, T3 800 steps (equals the env time-out)
success signal	terminated AND termination_manager.get_term('success'), evaluated per step
eval sets	frozen snapshots, batches 0-9 per level, never regenerated
total scored episodes	21,600 (plus 21,600 discarded warm-up episodes)

1.3 Training configuration, identical for all 108 cells

setting	value	setting	value
policy	LeRobot 0.4.4 diffusion policy	optimiser	Adam, lr 1e-4, wd 1e-6
training steps	80,000 (fixed, no early stop)	betas / grad clip	0.95, 0.999 / 10.0
seed	0 (one seed per cell)	scheduler	diffuser cosine, 500 warm-up steps
batch size	64	diffusion	DDPM, 100 train steps, squaredcos_cap_v2
vision backbone	ResNet-18, no pretrained weights	inference steps (eval)	10
RGB encoders	one per camera (2 cameras)	observation horizon	2 steps
cameras	table_cam, wrist_cam, 3x128x128	action horizon / exec	16 / 8
crop / norm	112x112, GroupNorm	state / action dim	9 / 7
normalisation	visual MEAN_STD, state+action MIN_MAX	norm statistics	full 400-episode pool, every budget

All 108 cells reached step 80,000 in a single Slurm allocation with no resume. The 126 resolved configuration keys are identical across the matrix apart from dataset path, episode list, job name, output directory and run id.

2 Data provenance

2.1 Demonstration generation and dataset size

task	arm	generation legs	demonstrations kept	failed attempts	attempts	generation SR	mean demo length (steps)	dataset episodes	dataset frames
T1 cup_place	L0	1	400	63	463	86.4 %	207	400	82,916
T1 cup_place	L1	1	400	66	466	85.8 %	188	400	75,366
T1 cup_place	L2	1	400	70	470	85.1 %	187	400	74,860
T1 cup_place	L3b	10	401	62	463	86.6 %	189	400	75,438
T1 cup_place	AC	10	400	49	449	89.1 %	188	400	75,294
T1 cup_place	BC	10	404	69	473	85.4 %	208	400	82,960
T2 drawer_stow	L0	1	400	328	728	54.9 %	705	400	281,987
T2 drawer_stow	L1	1	400	506	906	44.2 %	694	400	277,661
T2 drawer_stow	L2	1	400	906	1,306	30.6 %	677	400	270,744
T2 drawer_stow	L3b	10	400	947	1,347	29.7 %	674	400	269,650
T2 drawer_stow	AC	10	400	538	938	42.6 %	694	400	277,842
T2 drawer_stow	BC	10	400	1,002	1,402	28.5 %	680	400	271,895
T3 push_target	L0	1	400	6	406	98.5 %	317	400	126,735
T3 push_target	L1	1	400	22	422	94.8 %	319	400	127,609
T3 push_target	L2	1	400	21	421	95.0 %	311	400	124,472
T3 push_target	L3b	10	400	34	434	92.2 %	310	400	123,934
T3 push_target	AC	10	400	51	451	88.7 %	317	400	126,671
T3 push_target	BC	10	400	49	449	89.1 %	310	400	124,093

Every dataset holds exactly 400 episodes at 20 fps. Smaller budgets are contiguous prefixes of the same episode order, which is round-robin over the ten variants, so every budget of a variant arm is variant-balanced (N=10 gives 1 episode per variant, N=400 gives 40). BC in T1 produced 404 demonstrations across its ten legs; the four surplus episodes fall outside the 400-episode prefix and are unused.

Demonstration generation: yield per arm (400 demonstrations produced in every cell)

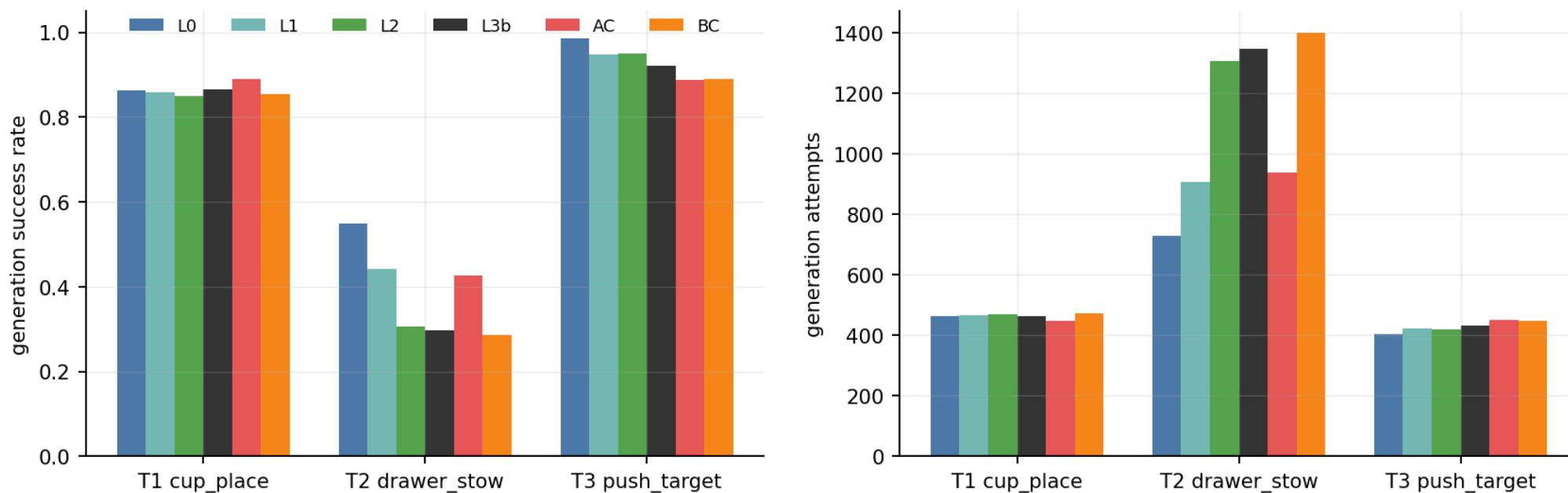


Figure 10. Demonstration yield per arm. Left: fraction of generation attempts that produced a usable demonstration. Right: attempts required. Every cell was filled to exactly 400 demonstrations.

3 Measured surface

All 108 cells, each 200 scored episodes. Intervals are Wilson score intervals at 95 % on the 200 episodes of that cell.

Measured surface: success rate vs demonstration budget, six arms per task

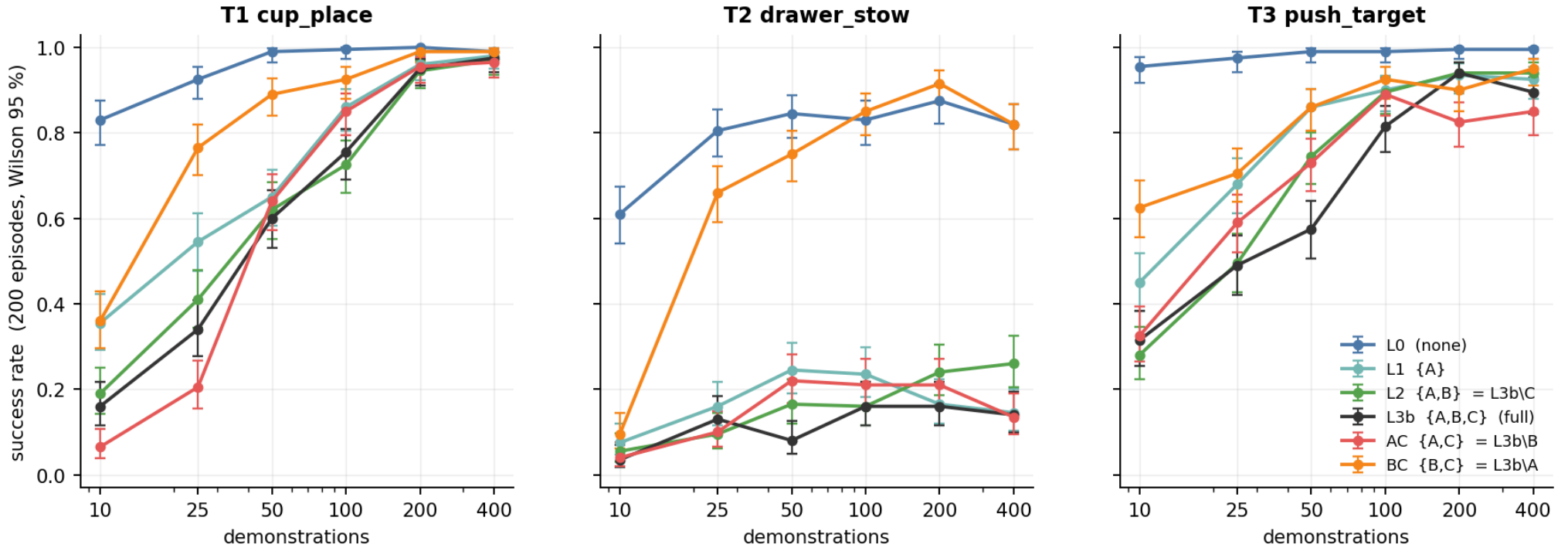


Figure 1. Success rate against demonstration budget for all six arms of each task. Error bars are Wilson 95 % intervals on 200 episodes.

Measured surface: 108 cells, 200 scored episodes each

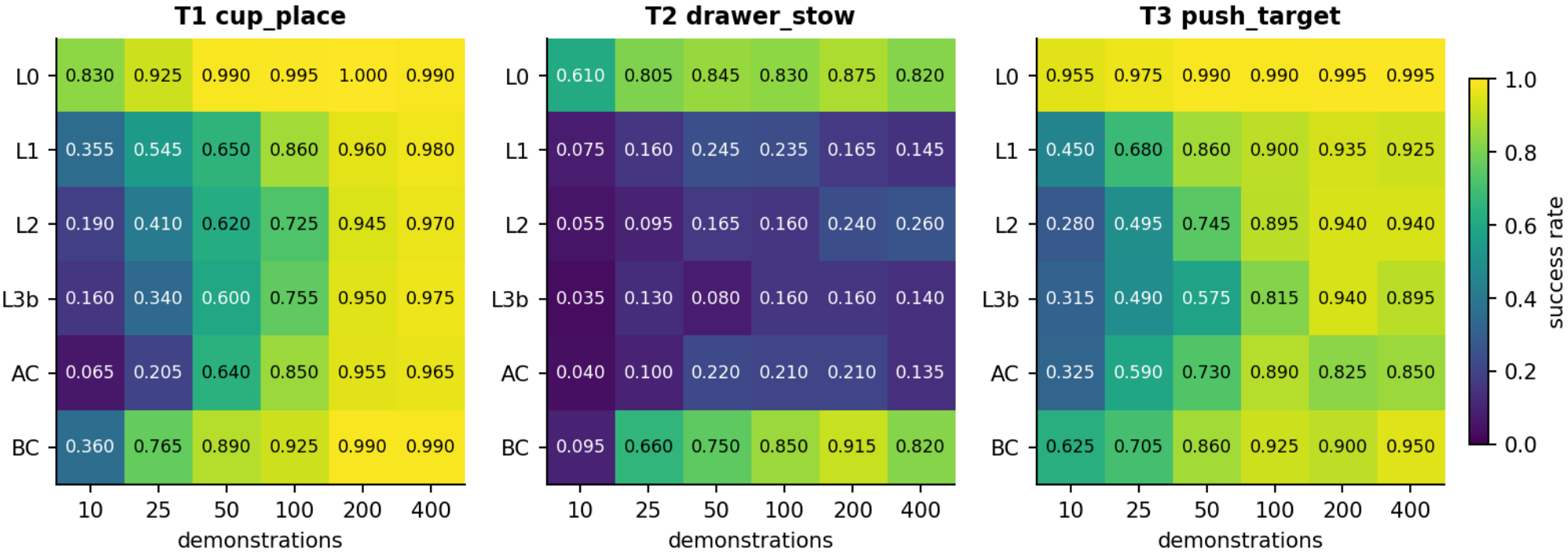


Figure 2. The same 108 measurements as a grid. Cell values are success rates over 200 episodes.

3.1 Cell table

T1 cup_place

arm	N=10	N=25	N=50	N=100	N=200	N=400
L0	0.830 166/200 [0.772, 0.876]	0.925 185/200 [0.880, 0.954]	0.990 198/200 [0.964, 0.997]	0.995 199/200 [0.972, 0.999]	1.000 200/200 [0.981, 1.000]	0.990 198/200 [0.964, 0.997]
L1	0.355 71/200 [0.292, 0.423]	0.545 109/200 [0.476, 0.613]	0.650 130/200 [0.582, 0.713]	0.860 172/200 [0.805, 0.901]	0.960 192/200 [0.923, 0.980]	0.980 196/200 [0.950, 0.992]
L2	0.190 38/200 [0.142, 0.250]	0.410 82/200 [0.344, 0.479]	0.620 124/200 [0.551, 0.684]	0.725 145/200 [0.659, 0.782]	0.945 189/200 [0.904, 0.969]	0.970 194/200 [0.936, 0.986]
L3b	0.160 32/200 [0.116, 0.217]	0.340 68/200 [0.278, 0.408]	0.600 120/200 [0.531, 0.665]	0.755 151/200 [0.691, 0.809]	0.950 190/200 [0.910, 0.973]	0.975 195/200 [0.943, 0.989]
AC	0.065 13/200 [0.038, 0.108]	0.205 41/200 [0.155, 0.266]	0.640 128/200 [0.571, 0.703]	0.850 170/200 [0.794, 0.893]	0.955 191/200 [0.917, 0.976]	0.965 193/200 [0.930, 0.983]
BC	0.360 72/200 [0.297, 0.429]	0.765 153/200 [0.702, 0.818]	0.890 178/200 [0.839, 0.926]	0.925 185/200 [0.880, 0.954]	0.990 198/200 [0.964, 0.997]	0.990 198/200 [0.964, 0.997]

T2 drawer_stow

arm	N=10	N=25	N=50	N=100	N=200	N=400
L0	0.610 122/200 [0.541, 0.675]	0.805 161/200 [0.745, 0.854]	0.845 169/200 [0.788, 0.889]	0.830 166/200 [0.772, 0.876]	0.875 175/200 [0.822, 0.914]	0.820 164/200 [0.761, 0.867]
L1	0.075 15/200 [0.046, 0.120]	0.160 32/200 [0.116, 0.217]	0.245 49/200 [0.191, 0.309]	0.235 47/200 [0.182, 0.298]	0.165 33/200 [0.120, 0.223]	0.145 29/200 [0.103, 0.200]
L2	0.055 11/200 [0.031, 0.096]	0.095 19/200 [0.062, 0.144]	0.165 33/200 [0.120, 0.223]	0.160 32/200 [0.116, 0.217]	0.240 48/200 [0.186, 0.304]	0.260 52/200 [0.204, 0.325]
L3b	0.035 7/200 [0.017, 0.070]	0.130 26/200 [0.090, 0.184]	0.080 16/200 [0.050, 0.126]	0.160 32/200 [0.116, 0.217]	0.160 32/200 [0.116, 0.217]	0.140 28/200 [0.099, 0.195]
AC	0.040 8/200 [0.020, 0.077]	0.100 20/200 [0.066, 0.149]	0.220 44/200 [0.168, 0.282]	0.210 42/200 [0.159, 0.272]	0.210 42/200 [0.159, 0.272]	0.135 27/200 [0.094, 0.189]
BC	0.095 19/200 [0.062, 0.144]	0.660 132/200 [0.592, 0.722]	0.750 150/200 [0.686, 0.805]	0.850 170/200 [0.794, 0.893]	0.915 183/200 [0.868, 0.946]	0.820 164/200 [0.761, 0.867]

T3 push_target

arm	N=10	N=25	N=50	N=100	N=200	N=400
L0	0.955 191/200 [0.917, 0.976]	0.975 195/200 [0.943, 0.989]	0.990 198/200 [0.964, 0.997]	0.990 198/200 [0.964, 0.997]	0.995 199/200 [0.972, 0.999]	0.995 199/200 [0.972, 0.999]
L1	0.450 90/200 [0.383, 0.519]	0.680 136/200 [0.612, 0.741]	0.860 172/200 [0.805, 0.901]	0.900 180/200 [0.851, 0.934]	0.935 187/200 [0.892, 0.962]	0.925 185/200 [0.880, 0.954]

arm	N=10	N=25	N=50	N=100	N=200	N=400
L2	0.280 56/200 [0.222, 0.346]	0.495 99/200 [0.426, 0.564]	0.745 149/200 [0.680, 0.800]	0.895 179/200 [0.845, 0.930]	0.940 188/200 [0.898, 0.965]	0.940 188/200 [0.898, 0.965]
L3b	0.315 63/200 [0.255, 0.382]	0.490 98/200 [0.422, 0.559]	0.575 115/200 [0.506, 0.641]	0.815 163/200 [0.755, 0.863]	0.940 188/200 [0.898, 0.965]	0.895 179/200 [0.845, 0.930]
AC	0.325 65/200 [0.264, 0.393]	0.590 118/200 [0.521, 0.656]	0.730 146/200 [0.665, 0.787]	0.890 178/200 [0.839, 0.926]	0.825 165/200 [0.766, 0.871]	0.850 170/200 [0.794, 0.893]
BC	0.625 125/200 [0.556, 0.689]	0.705 141/200 [0.638, 0.764]	0.860 172/200 [0.805, 0.901]	0.925 185/200 [0.880, 0.954]	0.900 180/200 [0.851, 0.934]	0.950 190/200 [0.910, 0.973]

4 Leave-one-out results

Each row removes one axis from the full set L3b and reports the change in success rate at a matched demonstration budget. Intervals are Newcombe 95 % intervals for the difference of two independent proportions, except where the two sides are episode-matched (axis C), where the paired McNemar statistic is given as well.

Leave-one-out: change in success rate when one axis is removed from the full set

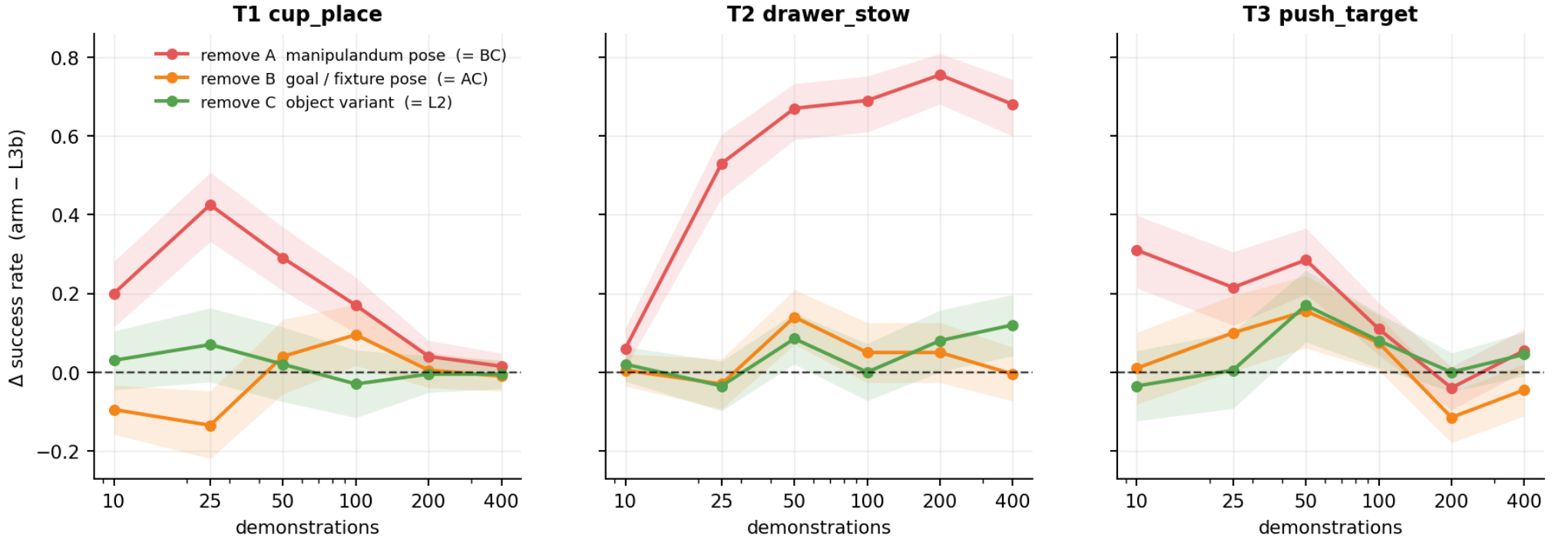


Figure 3. Leave-one-out delta against demonstration budget. Shaded bands are Newcombe 95 % intervals. The dashed line is no change.

Leave-one-out deltas, every task $\times$ axis $\times$ demonstration budget (faded = interval covers zero)

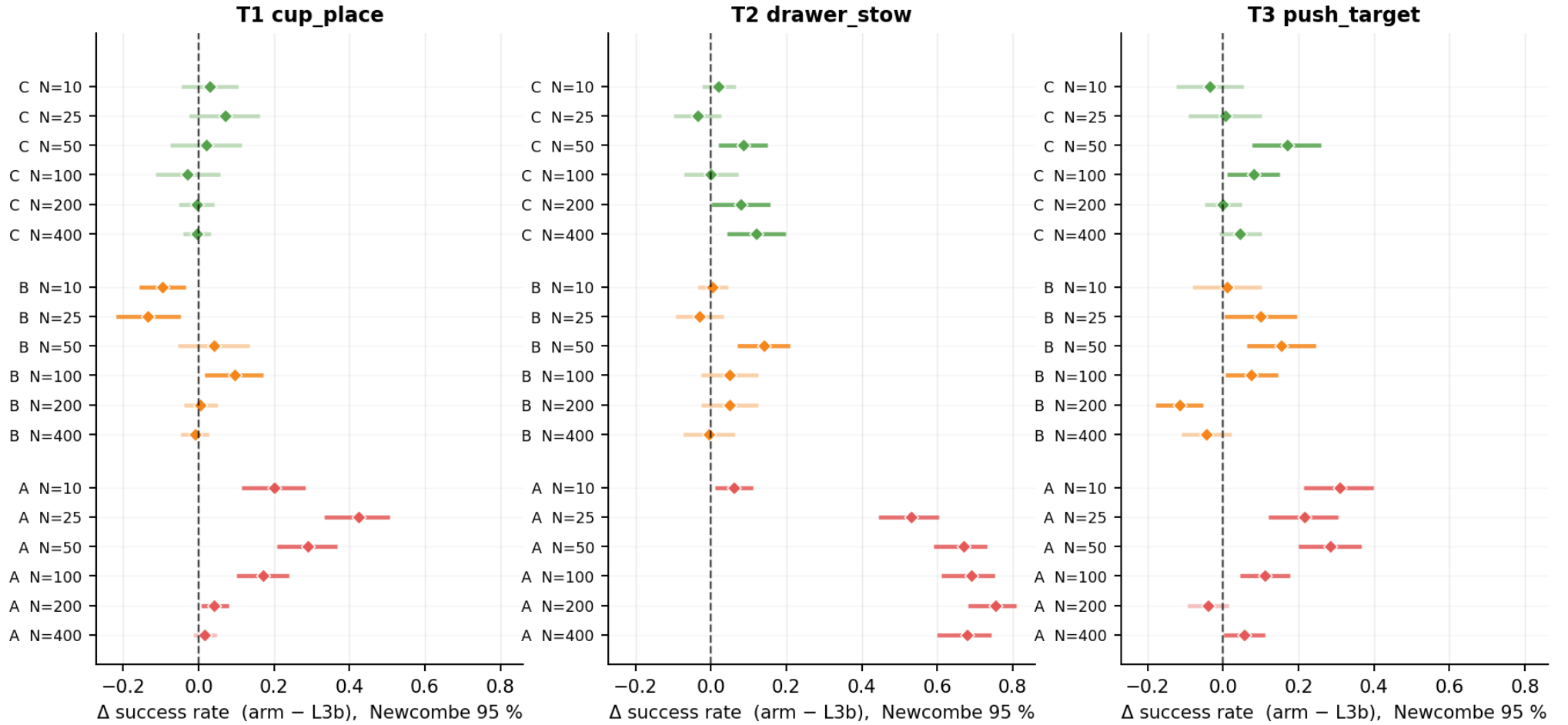


Figure 4. All 54 leave-one-out deltas with their intervals. Faded rows are those whose interval covers zero.

4.1 Delta table

task	axis removed	arm	N	SR L3b	SR reduced	delta	Newcombe 95 %	covers 0	paired McNemar (axis C only)
T1 cup_place	A	BC	10	0.160	0.360	+0.200	[+0.115, +0.282]	no	
T1 cup_place	A	BC	25	0.340	0.765	+0.425	[+0.332, +0.507]	no	
T1 cup_place	A	BC	50	0.600	0.890	+0.290	[+0.207, +0.368]	no	
T1 cup_place	A	BC	100	0.755	0.925	+0.170	[+0.099, +0.240]	no	
T1 cup_place	A	BC	200	0.950	0.990	+0.040	[+0.006, +0.080]	no	
T1 cup_place	A	BC	400	0.975	0.990	+0.015	[-0.014, +0.048]	yes	
T1 cup_place	B	AC	10	0.160	0.065	-0.095	[-0.158, -0.033]	no	
T1 cup_place	B	AC	25	0.340	0.205	-0.135	[-0.220, -0.048]	no	
T1 cup_place	B	AC	50	0.600	0.640	+0.040	[-0.055, +0.134]	yes	
T1 cup_place	B	AC	100	0.755	0.850	+0.095	[+0.017, +0.172]	no	
T1 cup_place	B	AC	200	0.950	0.955	+0.005	[-0.039, +0.050]	yes	
T1 cup_place	B	AC	400	0.975	0.965	-0.010	[-0.048, +0.027]	yes	
T1 cup_place	C	L2	10	0.160	0.190	+0.030	[-0.045, +0.105]	yes	b=27 c=21 d=+0.030 [-0.038, +0.098] chi2=0.52
T1 cup_place	C	L2	25	0.340	0.410	+0.070	[-0.025, +0.163]	yes	b=46 c=32 d=+0.070 [-0.016, +0.156] chi2=2.17
T1 cup_place	C	L2	50	0.600	0.620	+0.020	[-0.075, +0.115]	yes	b=42 c=38 d=+0.020 [-0.068, +0.108] chi2=0.11
T1 cup_place	C	L2	100	0.755	0.725	-0.030	[-0.115, +0.056]	yes	b=31 c=37 d=-0.030 [-0.111, +0.051] chi2=0.37
T1 cup_place	C	L2	200	0.950	0.945	-0.005	[-0.052, +0.041]	yes	b=9 c=10 d=-0.005 [-0.048, +0.038] chi2=0.00
T1 cup_place	C	L2	400	0.975	0.970	-0.005	[-0.042, +0.031]	yes	b=3 c=4 d=-0.005 [-0.031, +0.021] chi2=0.00
T2 drawer_stow	A	BC	10	0.035	0.095	+0.060	[+0.011, +0.112]	no	
T2 drawer_stow	A	BC	25	0.130	0.660	+0.530	[+0.443, +0.604]	no	
T2 drawer_stow	A	BC	50	0.080	0.750	+0.670	[+0.591, +0.733]	no	
T2 drawer_stow	A	BC	100	0.160	0.850	+0.690	[+0.610, +0.752]	no	
T2 drawer_stow	A	BC	200	0.160	0.915	+0.755	[+0.681, +0.809]	no	
T2 drawer_stow	A	BC	400	0.140	0.820	+0.680	[+0.599, +0.743]	no	
T2 drawer_stow	B	AC	10	0.035	0.040	+0.005	[-0.036, +0.046]	yes	
T2 drawer_stow	B	AC	25	0.130	0.100	-0.030	[-0.094, +0.033]	yes	
T2 drawer_stow	B	AC	50	0.080	0.220	+0.140	[+0.071, +0.209]	no	
T2 drawer_stow	B	AC	100	0.160	0.210	+0.050	[-0.026, +0.126]	yes	
T2 drawer_stow	B	AC	200	0.160	0.210	+0.050	[-0.026, +0.126]	yes	
T2 drawer_stow	B	AC	400	0.140	0.135	-0.005	[-0.073, +0.063]	yes	
T2 drawer_stow	C	L2	10	0.035	0.055	+0.020	[-0.023, +0.065]	yes	b=11 c=7 d=+0.020 [-0.021, +0.061] chi2=0.50
T2 drawer_stow	C	L2	25	0.130	0.095	-0.035	[-0.098, +0.028]	yes	b=14 c=21 d=-0.035 [-0.093, +0.023] chi2=1.03
T2 drawer_stow	C	L2	50	0.080	0.165	+0.085	[+0.021, +0.150]	no	b=30 c=13 d=+0.085 [+0.022, +0.148] chi2=5.95
T2 drawer_stow	C	L2	100	0.160	0.160	+0.000	[-0.072, +0.072]	yes	b=29 c=29 d=+0.000 [-0.075, +0.075] chi2=0.02
T2 drawer_stow	C	L2	200	0.160	0.240	+0.080	[+0.001, +0.158]	no	b=33 c=17 d=+0.080 [+0.012, +0.148] chi2=4.50

task	axis removed	arm	N	SR L3b	SR reduced	delta	Newcombe 95 %	covers 0	paired McNemar (axis C only)
T2 drawer_stow	C	L2	400	0.140	0.260	+0.120	[+0.042, +0.197]	no	b=38 c=14 d=+0.120 [+0.051, +0.189] chi2=10.17
T3 push_target	A	BC	10	0.315	0.625	+0.310	[+0.214, +0.398]	no	
T3 push_target	A	BC	25	0.490	0.705	+0.215	[+0.119, +0.305]	no	
T3 push_target	A	BC	50	0.575	0.860	+0.285	[+0.199, +0.366]	no	
T3 push_target	A	BC	100	0.815	0.925	+0.110	[+0.044, +0.176]	no	
T3 push_target	A	BC	200	0.940	0.900	-0.040	[-0.096, +0.014]	yes	
T3 push_target	A	BC	400	0.895	0.950	+0.055	[+0.002, +0.110]	no	
T3 push_target	B	AC	10	0.315	0.325	+0.010	[-0.081, +0.101]	yes	
T3 push_target	B	AC	25	0.490	0.590	+0.100	[+0.002, +0.195]	no	
T3 push_target	B	AC	50	0.575	0.730	+0.155	[+0.062, +0.245]	no	
T3 push_target	B	AC	100	0.815	0.890	+0.075	[+0.005, +0.145]	no	
T3 push_target	B	AC	200	0.940	0.825	-0.115	[-0.179, -0.052]	no	
T3 push_target	B	AC	400	0.895	0.850	-0.045	[-0.111, +0.021]	yes	
T3 push_target	C	L2	10	0.315	0.280	-0.035	[-0.124, +0.054]	yes	b=39 c=46 d=-0.035 [-0.125, +0.055] chi2=0.42
T3 push_target	C	L2	25	0.490	0.495	+0.005	[-0.092, +0.102]	yes	b=50 c=49 d=+0.005 [-0.093, +0.103] chi2=0.00
T3 push_target	C	L2	50	0.575	0.745	+0.170	[+0.077, +0.259]	no	b=62 c=28 d=+0.170 [+0.080, +0.260] chi2=12.10
T3 push_target	C	L2	100	0.815	0.895	+0.080	[+0.011, +0.149]	no	b=31 c=15 d=+0.080 [+0.014, +0.146] chi2=4.89
T3 push_target	C	L2	200	0.940	0.940	+0.000	[-0.049, +0.049]	yes	b=12 c=12 d=+0.000 [-0.048, +0.048] chi2=0.04
T3 push_target	C	L2	400	0.895	0.940	+0.045	[-0.010, +0.101]	yes	b=16 c=7 d=+0.045 [-0.002, +0.092] chi2=2.78

Shaded rows are those whose 95 % interval excludes zero.

5 Within-cell dispersion

Every cell is ten independent processes of 20 episodes. The spread of those ten slice rates is the sampling variability behind each pooled cell value.

Within-cell dispersion: the ten independent slices behind every pooled cell

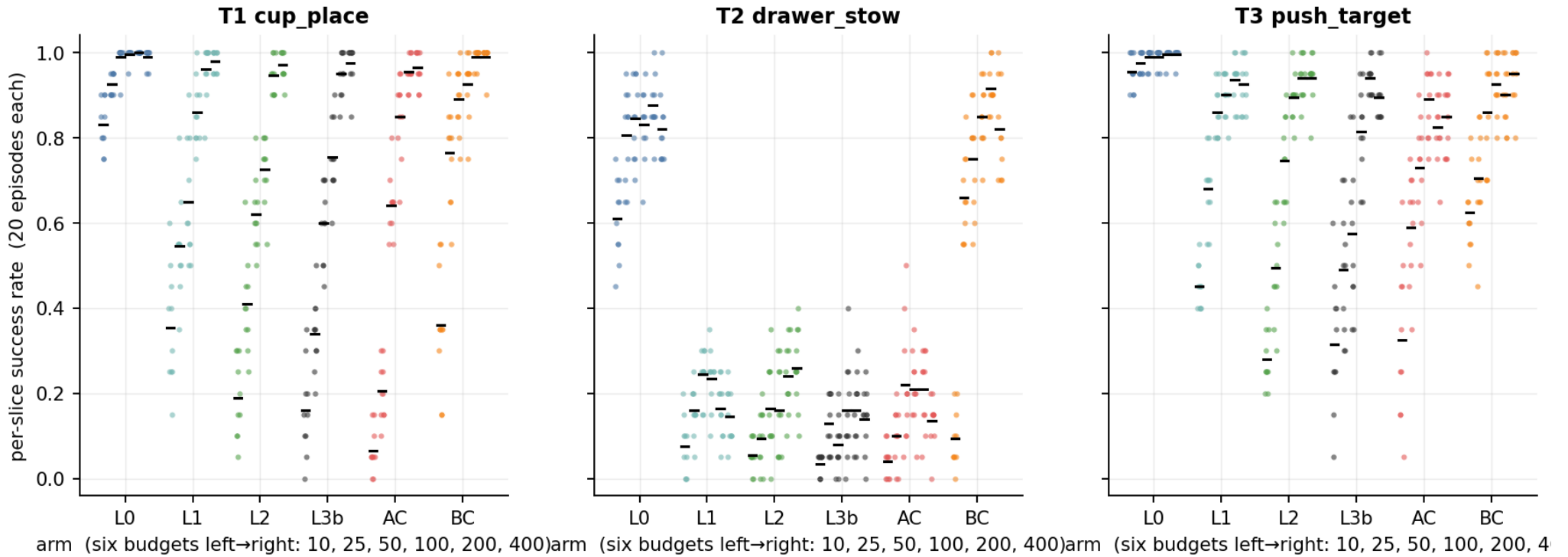


Figure 5. The ten slice success rates behind every cell (dots) and the pooled cell value (black bar).

5.1 Homogeneity of the ten slices within a cell

task	arm	N	pooled SR	slice min	slice max	chi2 (df 9)	rejects common rate at 95 %
T3 push_target	AC	10	0.325	0.05	0.65	28.8	yes
T3 push_target	L3b	50	0.575	0.35	0.85	25.5	yes
T3 push_target	BC	50	0.860	0.70	1.00	23.9	yes
T2 drawer_stow	AC	50	0.220	0.05	0.50	21.1	yes
T3 push_target	L3b	10	0.315	0.05	0.60	20.9	yes
T1 cup_place	L1	200	0.960	0.80	1.00	20.3	yes

task	arm	N	pooled SR	slice min	slice max	chi2 (df 9)	rejects common rate at 95 %
T2 drawer_stow	L3b	100	0.160	0.00	0.40	18.5	yes
T1 cup_place	BC	400	0.990	0.90	1.00	18.2	yes
T1 cup_place	L0	50	0.990	0.90	1.00	18.2	yes
T1 cup_place	L3b	400	0.975	0.85	1.00	17.4	yes
T3 push_target	BC	400	0.950	0.80	1.00	16.8	no
T3 push_target	L3b	25	0.490	0.30	0.70	16.3	no
T1 cup_place	BC	10	0.360	0.15	0.55	16.0	no
T1 cup_place	L1	50	0.650	0.50	0.90	15.8	no

10 of the 108 cells reject a common rate across their ten slices at 95 % (critical value 16.92 on 9 degrees of freedom). The fourteen largest statistics are listed.

6 T2 drawer_stow milestone instrumentation

T2 is the only task instrumented with intermediate milestones; T1 and T3 record binary success only. A milestone latches when its condition is met at any point in the episode: drawer opened past 0.15 m, object lifted past 0.05 m, object above the drawer cavity. All rates are over the same 200 episodes as the success rate.

T2 drawer_stow stage funnel: absolute rate of each milestone, per arm

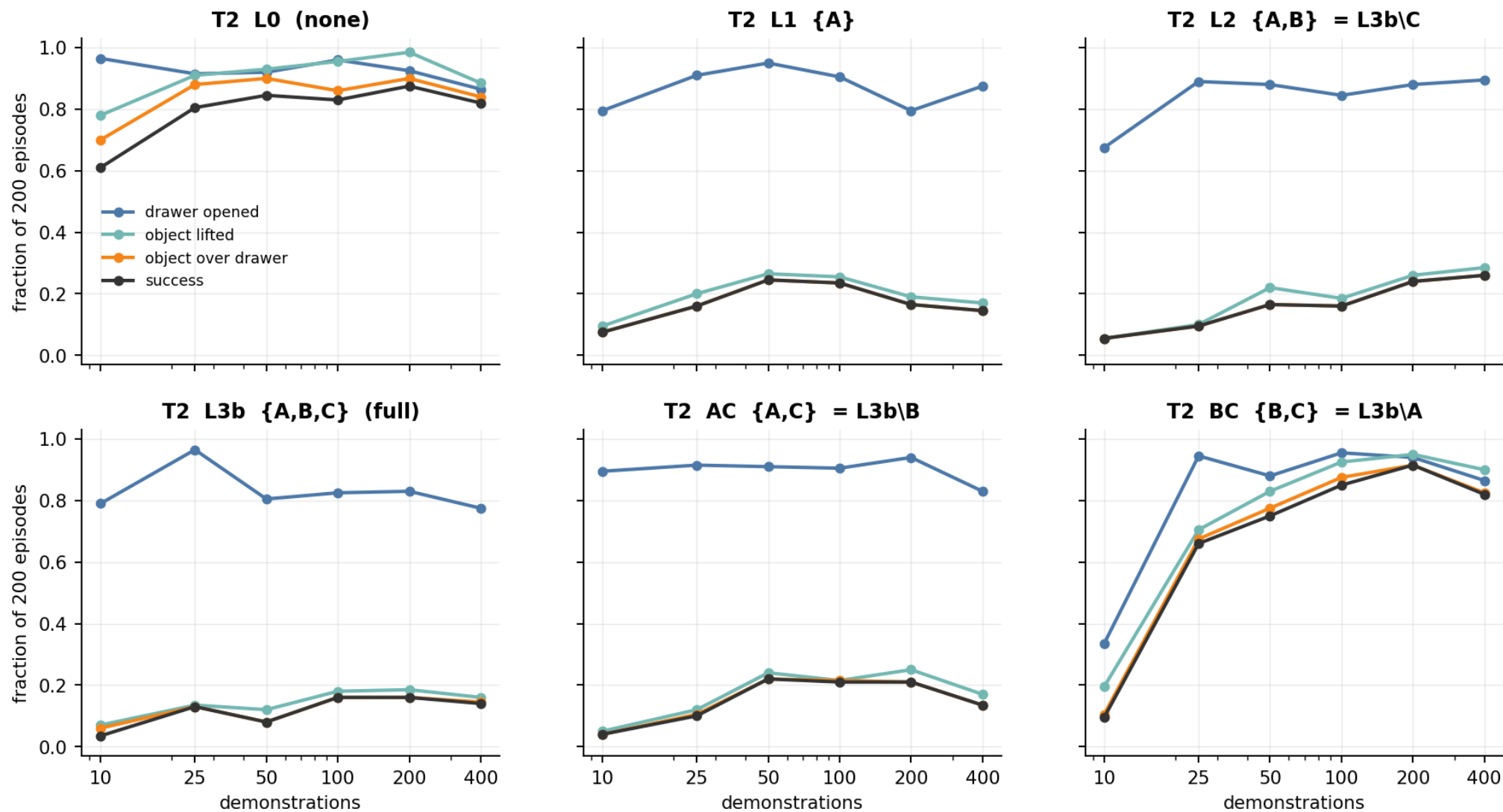


Figure 6. Absolute milestone rates per arm and budget, with success overlaid.

T2 drawer_stow: conditional transition rates between consecutive milestones

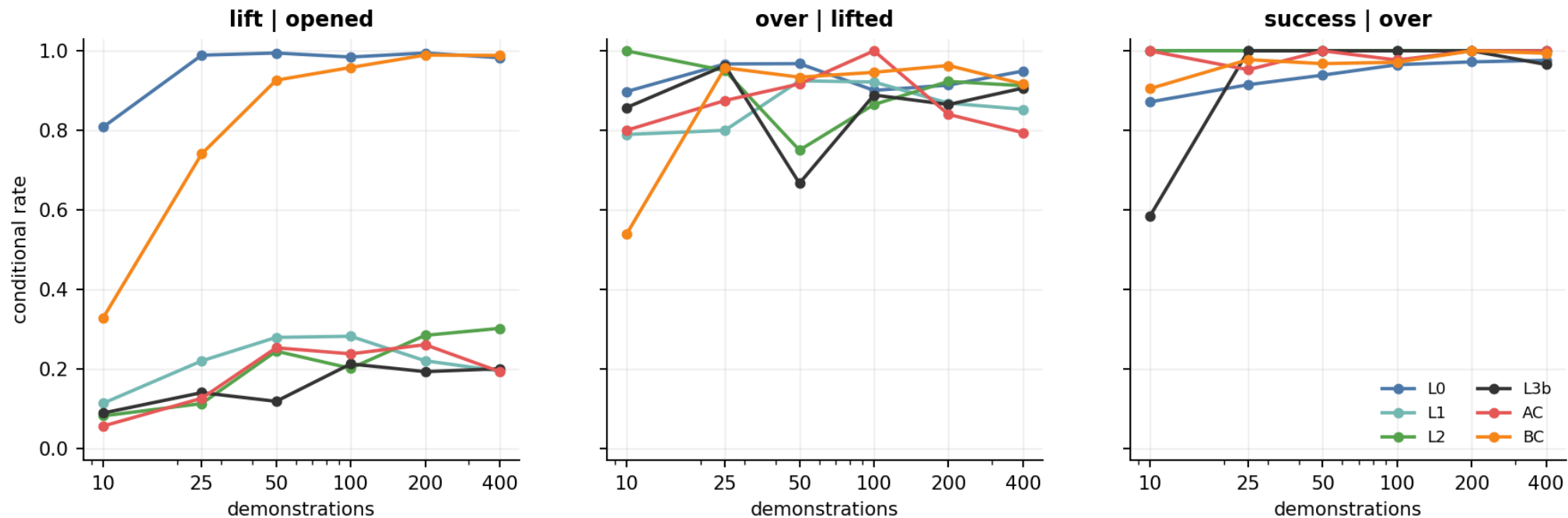


Figure 7. Conditional rate of reaching each milestone given the previous one.

T2 drawer_stow: continuous progress measures over all 1,200 episodes per arm

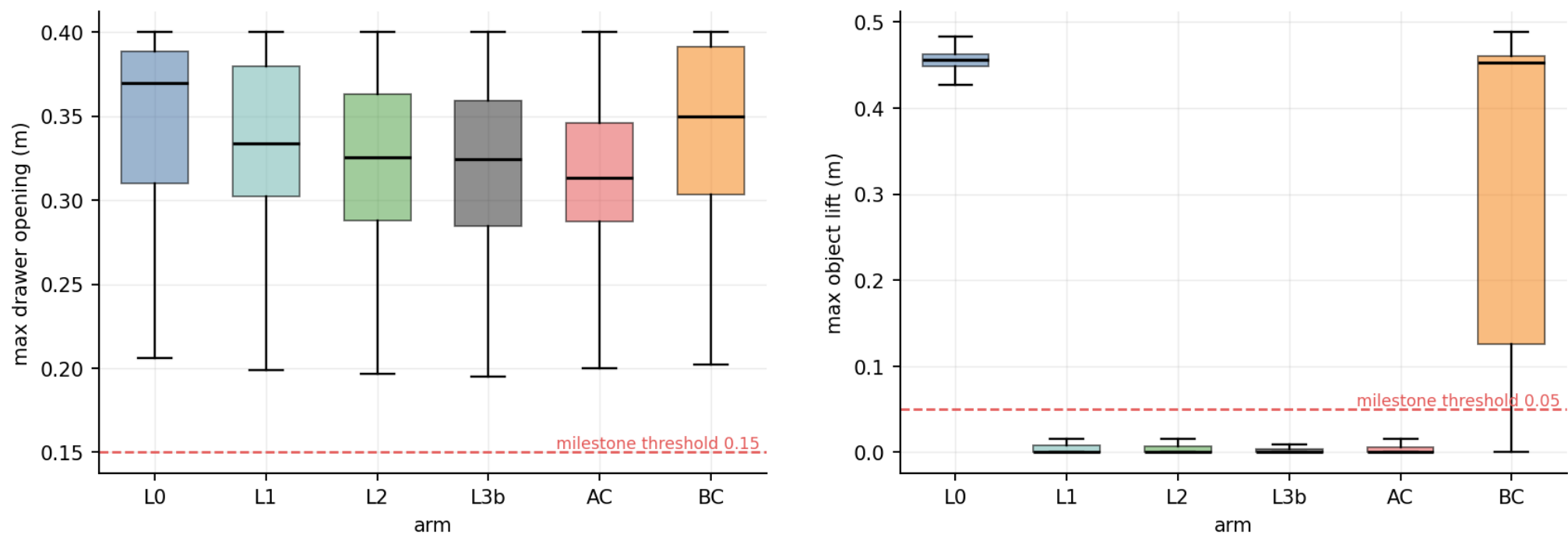


Figure 9. Continuous progress measures over all 1,200 episodes of each arm (six budgets pooled). Dashed lines mark the milestone thresholds.

The three milestones are not nested: an episode can lift the object without ever having opened the drawer past 0.15 m. Every conditional rate below is therefore computed from the joint count (episodes carrying both flags divided by episodes carrying the first), not from the ratio of two marginal rates. The joint counts themselves are in `paper/loo_report/tables/stages.csv`.

6.1 Milestone table

arm	N	drawer opened	object lifted	object over drawer	success	lift opened	over lifted	success over	median max opening (m)	median max lift (m)
L0	10	0.965	0.780	0.700	0.610	0.808	0.897	0.871	0.3888	0.4484
L0	25	0.915	0.910	0.880	0.805	0.989	0.967	0.915	0.3808	0.4565
L0	50	0.920	0.930	0.900	0.845	0.995	0.968	0.939	0.3694	0.4566
L0	100	0.960	0.955	0.860	0.830	0.984	0.901	0.965	0.3610	0.4567
L0	200	0.925	0.985	0.900	0.875	0.995	0.914	0.972	0.3454	0.4589
L0	400	0.865	0.885	0.840	0.820	0.983	0.949	0.976	0.3409	0.4573
L1	10	0.795	0.095	0.075	0.075	0.113	0.789	1.000	0.3398	0.0000
L1	25	0.910	0.200	0.160	0.160	0.220	0.800	1.000	0.3580	0.0006
L1	50	0.950	0.265	0.245	0.245	0.279	0.925	1.000	0.3395	0.0017
L1	100	0.905	0.255	0.235	0.235	0.282	0.922	1.000	0.3379	0.0010
L1	200	0.795	0.190	0.165	0.165	0.220	0.868	1.000	0.3106	0.0000
L1	400	0.875	0.170	0.145	0.145	0.194	0.853	1.000	0.3233	0.0000
L2	10	0.675	0.055	0.055	0.055	0.081	1.000	1.000	0.3177	0.0000
L2	25	0.890	0.100	0.095	0.095	0.112	0.950	1.000	0.3396	0.0000
L2	50	0.880	0.220	0.165	0.165	0.244	0.750	1.000	0.3321	0.0001
L2	100	0.845	0.185	0.160	0.160	0.201	0.865	1.000	0.3218	0.0000
L2	200	0.880	0.260	0.240	0.240	0.284	0.923	1.000	0.3242	0.0001
L2	400	0.895	0.285	0.260	0.260	0.302	0.912	1.000	0.3191	0.0008
L3b	10	0.790	0.070	0.060	0.035	0.089	0.857	0.583	0.3236	0.0000
L3b	25	0.965	0.135	0.130	0.130	0.140	0.963	1.000	0.3488	0.0000
L3b	50	0.805	0.120	0.080	0.080	0.118	0.667	1.000	0.3289	0.0000
L3b	100	0.825	0.180	0.160	0.160	0.212	0.889	1.000	0.3132	0.0002
L3b	200	0.830	0.185	0.160	0.160	0.193	0.865	1.000	0.3156	0.0000
L3b	400	0.775	0.160	0.145	0.140	0.200	0.906	0.966	0.3121	0.0000
AC	10	0.895	0.050	0.040	0.040	0.056	0.800	1.000	0.2898	0.0000
AC	25	0.915	0.120	0.105	0.100	0.126	0.875	0.952	0.3043	0.0000
AC	50	0.910	0.240	0.220	0.220	0.253	0.917	1.000	0.3231	0.0000
AC	100	0.905	0.215	0.215	0.210	0.238	1.000	0.977	0.3261	0.0000
AC	200	0.940	0.250	0.210	0.210	0.261	0.840	1.000	0.3216	0.0000
AC	400	0.830	0.170	0.135	0.135	0.193	0.794	1.000	0.3175	0.0000
BC	10	0.335	0.195	0.105	0.095	0.328	0.538	0.905	0.0000	0.0000
BC	25	0.945	0.705	0.675	0.660	0.741	0.957	0.978	0.3526	0.4563
BC	50	0.880	0.830	0.775	0.750	0.926	0.934	0.968	0.3676	0.4541

arm	N	drawer opened	object lifted	object over drawer	success	lift opened	over lifted	success over	median max opening (m)	median max lift (m)
BC	100	0.955	0.925	0.875	0.850	0.958	0.946	0.971	0.3638	0.4517
BC	200	0.940	0.950	0.915	0.915	0.989	0.963	1.000	0.3594	0.4539
BC	400	0.865	0.900	0.825	0.820	0.988	0.917	0.994	0.3449	0.4552

7 Time to success

Step index at which a successful episode first satisfied the success condition, over successful episodes only, all budgets pooled.

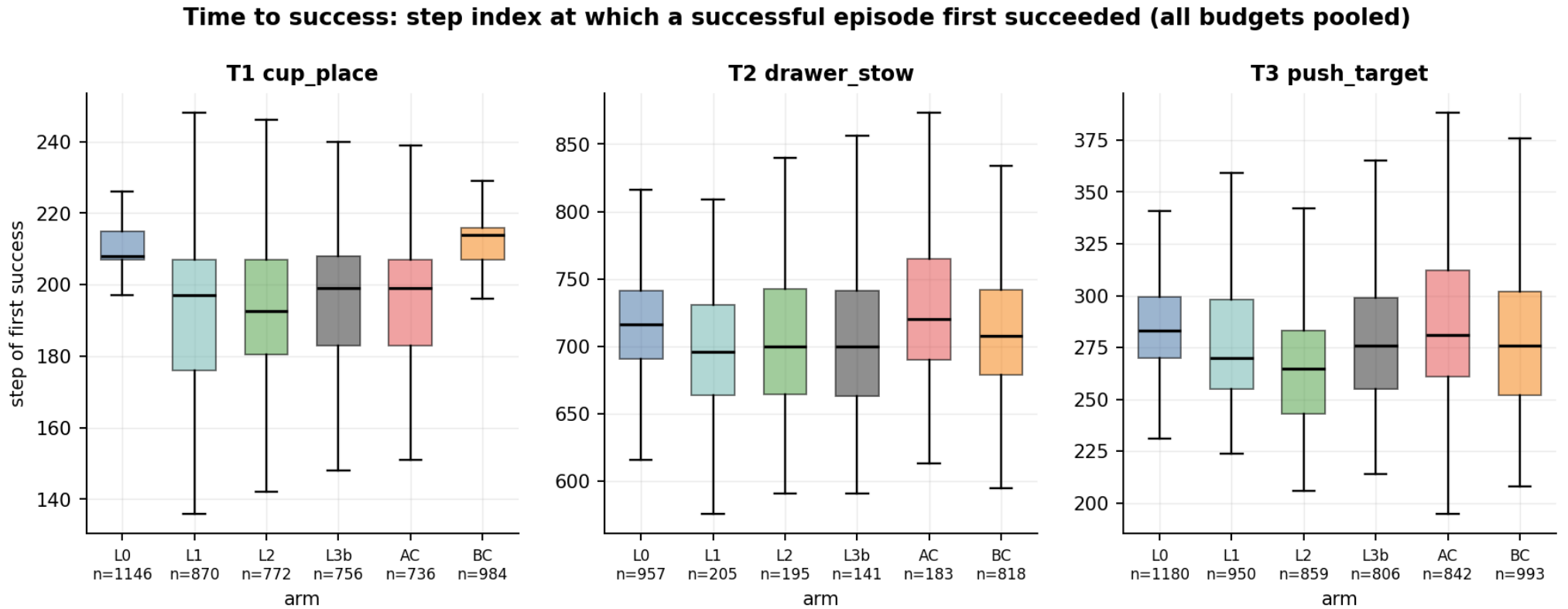


Figure 8. Distribution of the first-success step per arm. Boxes are quartiles, whiskers 1.5 IQR, outliers suppressed.

task	arm	successful episodes	min	p25	median	p75	max	episode cap
T1 cup_place	L0	1,146	191	207	208	215	512	600
T1 cup_place	L1	870	136	176	197	207	488	600
T1 cup_place	L2	772	123	180	192	207	383	600
T1 cup_place	L3b	756	144	183	199	208	496	600
T1 cup_place	AC	736	137	183	199	207	550	600
T1 cup_place	BC	984	184	207	214	216	509	600
T2 drawer_stow	L0	957	613	691	716	741	1181	1200
T2 drawer_stow	L1	205	576	664	696	731	882	1200

task	arm	successful episodes	min	p25	median	p75	max	episode cap
T2 drawer_stow	L2	195	591	664	700	742	1041	1200
T2 drawer_stow	L3b	141	591	663	700	741	995	1200
T2 drawer_stow	AC	183	613	690	720	765	1057	1200
T2 drawer_stow	BC	818	595	679	708	742	1186	1200
T3 push_target	L0	1,180	231	270	283	299	341	800
T3 push_target	L1	950	224	255	270	298	661	800
T3 push_target	L2	859	206	243	265	283	617	800
T3 push_target	L3b	806	214	255	276	299	789	800
T3 push_target	AC	842	195	261	281	312	679	800
T3 push_target	BC	993	141	252	276	302	638	800

8 Implementation flaws

Each entry is a property of how this study was implemented or measured, with the quantity that establishes it. They are listed, not weighted.

id	flaw	evidence
F1	Episode-level eval is not bit-reproducible	Re-running one slice with an identical environment, seed, checkpoint, warm-up and code reproduced the slice total (16/20) but flipped two individual environments, and the first-success step differed on 14 of 20. Four of five independently re-run verification slices differ from their originals by one success (17 vs 16, 17 vs 16, 5 vs 4, 17 vs 18). Per-cell totals are therefore reproducible only up to binomial-scale noise of roughly +/-1 success per 20-episode slice; no single slice is an exact regression test.
F2	One training seed per cell	Every cell is seed 0. All intervals in this report are episode-sampling intervals only; they contain no training-seed variance, which is unmeasured.
F3	Demonstrations for AC and BC were generated on different GPUs than L0-L3b	AC and BC demonstrations were generated on A100-SXM-64GB (driver 535.274.02); L0, L1, L2 and L3b demonstrations were generated on an RTX 4090 (driver 580.173.02). Training and evaluation are common to all arms (same A100 partition, identical configuration). The A and B leave-one-out rows therefore compare arms whose demonstrations were rendered on different hardware; the C row does not (L2 and L3b are both 4090). A dedicated control cell (L1 demonstrations regenerated on A100, seed 1100, retrained, re-evaluated under this report's protocol) scored 0.750 [100 episodes] against the 4090-generated L1 cell's 0.860 [200 episodes]; difference -0.110, 95% CI [-0.208, -0.012]. That control varies both the GPU and the demonstration RNG stream, so it bounds rather than isolates the effect.
F4	The C arm is not an independently generated arm	L3b \ C is bit-identical to L2 by construction: L2's fixed object is a member of the variant set, and the two environments were verified to produce identical initial states over 300 matched resets in T1 and T3 and identical configuration fields in all three tasks. The C row therefore re-uses the existing L2 cells rather than a separately generated arm, and is the only row whose two sides are episode-matched.
F5	Pooling across demonstration budgets is statistically invalid	The six budgets within one arm are six different policies on a rising curve. A homogeneity test across the six cells of an arm rejects a common proportion in most arms (see Appendix A). The budget-pooled deltas in Appendix A are reported only with their homogeneity statistic beside them; the per-budget rows in Section 4 are the primary result.
F6	Residual first-batch depression after the warm-up batch is unmeasured	One unscored warm-up batch is run in every slice of every cell, so the setting cannot differ between arms. Measured on one cell, the scored success rate rises from 0.72 with no warm-up to 0.85 with one full warm-up batch. Whether a second warm-up batch would raise it further has not been measured. One structural asymmetry remains: flat arms warm up in the same sub-environment in all ten slices, variant arms in a different one each.
F7	The pooling guard cannot detect a mixed-generation pool	scripts/ops/pool_variant_eval.py keys its MIXED_PROTOCOL check on (num_inference_steps, warm-up block), which is identical across all three evaluator generations produced during this study. The two fields that do differ (success_signal, phantom_guard_steps) are not checked, and the pooled payload drops both, so a pooled file's only provenance is its filename suffix. All 108 cells in this report were verified slice-by-slice to carry the corrected success signal; no mixed pool occurred.
F8	The corrected success signal depends on an unasserted term ordering	The per-step success flag is exact because 'success' is the last declared termination term in all three task configurations, so it always wins the manager's one-hot record. Nothing asserts this. Appending a termination term after 'success' would silently begin dropping successes.
F9	T2 milestone flags are evaluated one step after the auto-reset boundary	For the single step on which an environment terminates, the milestone tensors are read after the vectorised environment has already reset that row. Measured over all 7,180 T2 episodes the effect is not observable: zero episodes carry drawer_opened with a maximum opening below threshold, zero carry object_lifted below threshold, and zero successes are missing a milestone.
F10	Provenance gaps in the ledgers	No git revision is recorded in any run artefact (the cluster checkout carries no .git). Registry rows do not record which evaluator generation produced their success rate. Three development scripts still read the raw termination latch without the per-step guard (scripts/dev/t2_smoke.py:57, t3_smoke.py:81, sm_diag.py:69); no number in this report comes from them. Four T1 BC generation legs overshot to 41 demonstrations each (trimmed to 400 at conversion). A failed first eval-set freeze wave remains in experiments/cluster_jobs.csv beside the successful one, with no status column.
F11	A docstring claim that does not hold for BC	pool_variant_eval.py and slurm/eval.sbatch state that every arm has the same spatial coverage. BC covers 20 distinct manipulandum XY poses across its 200 episodes rather than 200, because axis A is frozen. This is by construction, not a measurement error.

9 Open items

id	item	status
T1	Assert the termination-term ordering that F8 depends on, in the eval entry point.	open
T2	Extend the pooling guard to key on success_signal and phantom_guard_steps, and carry both into the pooled payload.	open
T3	Write the evaluator generation into experiments/registry.csv rows and notes.	open
T4	Measure the residual first-batch effect with a two-warm-up-batch slice, flat and variant arm.	open
T5	Re-run the GPU-portability control with the demonstration RNG stream held fixed, so the GPU is the only varying factor (F3).	open
T6	Record a git revision in every run artefact.	open
T7	Add the per-step guard to the three development scripts named in F10.	open
T8	Correct the spatial-coverage docstrings for BC (F11).	open
T9	Merge the AC/BC generation rows into experiments/gen_stats.csv.	done 2026-09-20
T10	Move the leave-one-out computation out of a scratch script into the repository (src/cog/analysis/loo.py).	done 2026-09-20

Appendix A Budget-pooled deltas

The same deltas with all six budgets summed into one 1,200-episode proportion per side. This pooling is reported for completeness only: the six budgets are six different policies, and the homogeneity statistic beside each row tests whether they share one success rate (critical value 11.07 on 5 degrees of freedom). Where that test rejects, the pooled interval describes a mixture whose value depends on which budgets were included, not a single parameter.

task	axis removed	arm	successes L3b	successes reduced	episodes/side	pooled delta	Newcombe 95 %	chi2 L3b (df 5)	chi2 reduced (df 5)	homogeneity rejected
T1 cup_place	A	BC	756	984	1,200	+0.190	[+0.155, +0.225]	465.9	390.7	yes
T1 cup_place	B	AC	756	736	1,200	-0.017	[-0.055, +0.022]	465.9	644.8	yes
T1 cup_place	C	L2	756	772	1,200	+0.013	[-0.025, +0.052]	465.9	405.2	yes
T2 drawer_stow	A	BC	141	818	1,200	+0.564	[+0.531, +0.595]	24.1	415.9	yes
T2 drawer_stow	B	AC	141	183	1,200	+0.035	[+0.008, +0.062]	24.1	41.6	yes
T2 drawer_stow	C	L2	141	195	1,200	+0.045	[+0.017, +0.073]	24.1	46.5	yes
T3 push_target	A	BC	806	993	1,200	+0.156	[+0.122, +0.190]	282.9	121.7	yes
T3 push_target	B	AC	806	842	1,200	+0.030	[-0.007, +0.067]	282.9	217.7	yes
T3 push_target	C	L2	806	859	1,200	+0.044	[+0.007, +0.081]	282.9	365.9	yes

Appendix B Files behind this report

path	contents
results/eval_<TASK>_<ARM>_n<N>_080000_u200d32.json	the 108 pooled cells, each carrying its 200 per-episode outcome records
paper/loo_report/tables/surface.csv	108 cells: successes, Wilson interval, slice spread, homogeneity
paper/loo_report/tables/loo.csv	54 leave-one-out rows with Newcombe and McNemar statistics
paper/loo_report/tables/pooled.csv	9 budget-pooled rows with homogeneity statistics
paper/loo_report/tables/stages.csv	36 T2 milestone rows
paper/loo_report/tables/timing.csv	18 first-success timing summaries
paper/loo_report/tables/gen.csv	18 generation-yield rows
src/cog/analysis/loo.py	every statistic in this report
src/cog/analysis/loo_figures.py	every figure in this report
scripts/dev/build_loo_report.py	this document